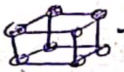
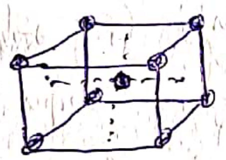
Characteristics of a Unit Cell :-

1. **Number of atoms per unit Cell (n) :-** it is the number of atoms present in a unit cell. This may be defined as total number of atoms present in (or) shared by an unit cell.

We shall calculate the no of atoms in the following cases:

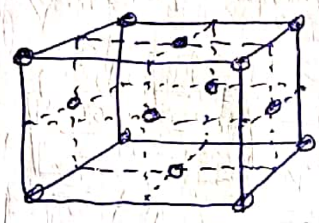
(i) **Simple cubic structure (S.C.) :-** having lattice points only at the corners of a unit cell.  – S.C.

(ii) **Body centred Structure (B.C.C) :-** having lattice points at the corners as well as at the body centre of the unit cell.



⟹ B.C.C

(iii) **Face centred cubic (F.C.C) :-** having lattice points at the corners as well as at the face centres of the unit cell.



⟹ F.C.C.

(1.) **Number of atoms per unit cell in S.C. Structure :-**

We know that there is one lattice point (atom) at each of the eight corners of the unit cell. Moreover, each lattice point forming the simple cubic lattice is a member of 8 surrounding cells. Therefore, the share of each lattice point to the cubic lattice = $1/8$, Number of lattice point in unit cell = $\frac{1}{8} \times 8 = \underline{1}$

In this way, there is only one lattice point per unit cell.

(2) **No of atoms per unit cell in BCC structure:**

No of corner point attached to unit cell = $\frac{1}{8} \times 8 = 1$

Also, there is one lattice point at the centre of the body.

Therefore, total no of BCC lattice point per unit cell = $1+1 = 2$

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(3) No of atoms per unit cell in a FCC structure:-

Total no of corner lattice point concerned with unit cell $= 8 \times \frac{1}{8} = 1$

Moreover, there exists six face lattice points, each being a member of two cells.

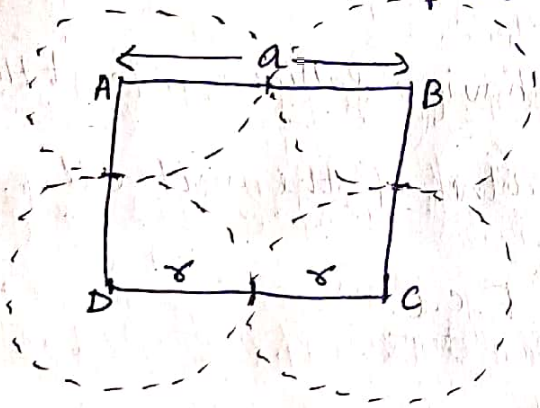
Total no of face centred lattice points concerned with unit cell $= 6 \times \frac{1}{2} = 3$

Hence, total no of lattice points concerned with unit cell $= 1 + 3 = \underline{4}$

2. Atomic Radius :- This is defined as half of the centre to centre distance of neighbouring atoms in a crystal. This is denoted by r. it is usually expressed in terms of cube edge a. The atomic radius $r = \frac{a}{2}$.

(1) Atomic radius for simple cubic (SC) structure

The front view of SC lattice is shown in fig:-

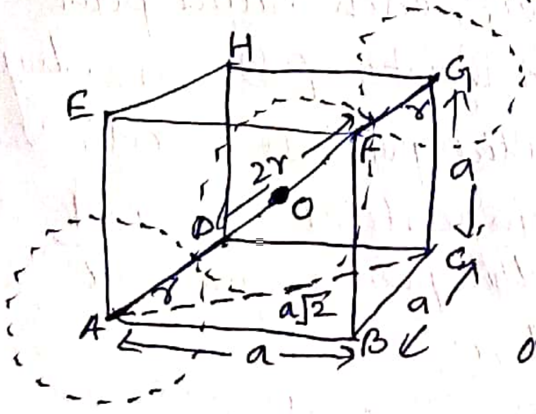


it is clear from fig that the distance b/w the centres of two nearest atoms in just equal to cube edge a, Therefore

$$2r = a$$

or, $$\boxed{r = \frac{a}{2}}$$

(2) Atomic radius for BCC structure: In case of BCC, the corner atoms do not touch each other. However, each corner atom touches the central atom as shown in fig below:-



For calculation of atomic radius, from fig:

$$(AG)^2 = (AC)^2 + (CG)^2 \quad \text{——— (1)}$$

(From ΔACG)

But $$(AC)^2 = (AB)^2 + (BC)^2 \quad \text{——— (2)}$$

(From ΔABC)

or, $(AC)^2 = a^2 + b^2 = 2a^2$, From eq^n(1), $(AG)^2 = 2a^2 + a^2 = 3a^2$

or $(4r)^2 = 3a^2 \Rightarrow \boxed{r = \frac{\sqrt{3}a}{4}}$

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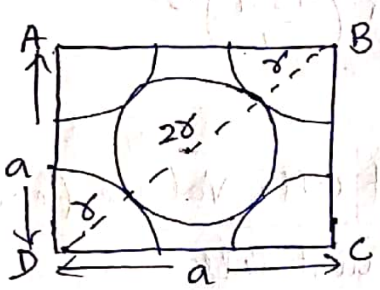
(3) Atomic radius for FCC structure :- The atoms touch each other along the diagonal of any face of the cube. The length of diagonal of the face is $4r$. The front view of FCC is shown below:



From Fig: $(DB)^2 = (DC)^2 + (CB)^2$

or, $(4r)^2 = a^2 + a^2 = 2a^2$

or, $4r = \sqrt{2}\,a$

$$\boxed{r = \frac{\sqrt{2}\,a}{4}}$$

3. Packing Factor (PF) :- The atomic packing factor is defined as the ratio of total volume occupied by the atoms in a unit cell (v) to the total volume of unit cell (V). Therefore,

$$\text{Packing factor} = \frac{\text{Total volume occupied by atoms in unit cell } (v)}{\text{Total volume of unit cell}}$$

This may be defined as:

$$\boxed{\text{Packing factor} = \frac{\text{No of atoms present in unit cell} \times \text{Volume of one atom}}{\text{Total Volume of unit cell}}}$$

The packing factor tells us how closely the atoms are stacked in a unit cell.

When packing factor is high, then atoms are closely packed. On the other hand, when packing factor is low, then atoms are loosely packed.

(1) APF for SC structure: In S.C. No of atoms per unit cell = 1

Atomic radius $r = \frac{a}{2}$

Volume of one atom $v = \frac{4}{3} \times \pi \times r^3 = \frac{4}{3}\pi \times \left(\frac{a}{2}\right)^3$ and $V = a^3$

Then, $P.F. = \frac{\text{No of atoms Present in unit cell} \times \text{Volume of one atom}}{\text{Total volume of unit cell}}$

$$P.F = \frac{1 \times \frac{4}{3}\pi\left(\frac{a}{2}\right)^3}{a^3} = \frac{4\pi a^3}{24\, a^3} = \frac{\pi}{6} = \frac{3.14}{6} = 0.5236 = \underline{\underline{52\%}}$$

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(2) Packing factor for BCC structure :- We know that in case of BCC,

No of atoms per unit cell = 2

Atomic radius $r = \frac{\sqrt{3}\,a}{4}$, Unit cell length = a

Volume of one atom $v = \frac{4}{3}\pi r^3 = \frac{4}{3}\pi\left(\frac{\sqrt{3}\,a}{4}\right)^3$

Total Volume of unit cell $V = a^3$

We know that :-

$$P.F. = \frac{\text{No of atoms present in unit cell} \times \text{Volume of one atom}}{\text{Total volume of unit cell}}$$

or $$P.F. = \frac{2 \times \frac{4}{3}\pi\left(\frac{\sqrt{3}\,a}{4}\right)^3}{a^3} = \frac{8\pi(\sqrt{3})^2 a^3}{3 \times 64 \times a^3} = \frac{\sqrt{3}\,\pi}{8} = 0.68$$

$\therefore$ $\boxed{P.F. = 68\%}$

(3) Packing factor for FCC Structure :- We know that in case of FCC,

No. of atoms per unit cell = 4

radius of each atom = $\left(\frac{\sqrt{2}\,a}{4}\right)$

$\therefore$ Volume of each atom $= \frac{4}{3}\pi r^3 = \frac{4}{3}\pi\left[\frac{\sqrt{2}\,a}{4}\right]^3 = \frac{\pi a^3\sqrt{2}}{24}$

Total Volume of unit cell = a^3

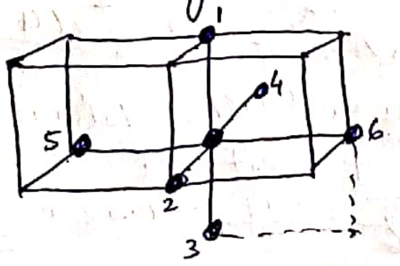
Then $$P.F. = \frac{\text{No of atoms present in unit cell} \times \text{Vol. of each atom}}{\text{Total volume of each cell}}$$

$$P.F. = \frac{4 \times (\pi a^3\sqrt{2})/24}{a^3} = \frac{\pi\sqrt{2}}{6} = 0.74$$

$\boxed{P.F. = 74\%}$

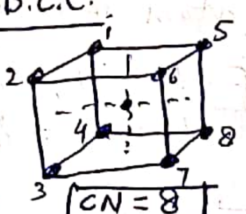
4. Co-ordination number (CN) : The Coordination number is defined as the number of equidistant neighbours around an atom or lattice points in the crystal lattice.

(i) S.C.



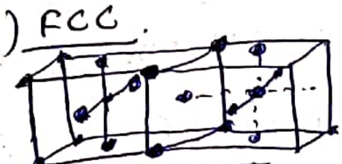
$\boxed{CN = 6}$

(ii) B.C.C.



$\boxed{CN = 8}$

(iii) FCC



$\boxed{CN = 12}$

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Numerical Questions :-

Ques-1: Lattice Constant of a BCC crystal is 0.36 nm. Calculate its atomic radius.

Soln:- Given $a = 0.36$ nm.

In case of BCC crystal, $r = \frac{a\sqrt{3}}{4} = \frac{0.36 \times \sqrt{3}}{4} = 0.16$ nm

Ques-2: Copper is FCC whose atomic radius is 1.26×10^{-19} m. Calculate its lattice Constant.

Soln: Given $r = 1.26 \times 10^{-10}$ m.

In case of FCC crystal, $r = \frac{\sqrt{2}a}{4} \Rightarrow a = \frac{4r}{\sqrt{2}}$

$a = \frac{4 \times (1.26 \times 10^{-10})}{\sqrt{2}} = 3.56 \times 10^{-10}$ m $\Rightarrow \boxed{a = 3.56 \times 10^{-10} m}$

Ques-3: Iron has BCC structure with atomic radius 0.123 A°. Find the volum of cell.

Soln: In Case of BCC, $r = \frac{a\sqrt{3}}{4}$ or $a = \frac{4r}{\sqrt{3}} = \frac{4 \times 0.123 \times 10^{-10}}{\sqrt{3}}$

$a = 0.284 \times 10^{-10}$ m

Volume of unit Cell $= V = a^3 = (0.284 \times 10^{-10})^3$

$\boxed{V = 2.29 \times 10^{-32} m^3}$

Ques-4:- Metallic iron changes from BCC to FCC form at 910°C and Correspondingly atomic radii vary from 1.258 A° to 1.292 A°. Calculate the percent volume change during this structural change.

Soln:- For BCC structure - $a = \frac{4r}{\sqrt{3}} = \frac{4 \times 1.258}{\sqrt{3}} = 2.9$ A°

$\therefore$ Volume of unit cell of BCC $= a^3 = 24.52 \times 10^{-30} m^3$

No of atoms per unit cell = 2

Therefore volume occupied by one atom $= \frac{24.52 \times 10^{-30}}{2}$

$v = 12.26 \times 10^{-30} m^3$

For FCC structure $a = 2\sqrt{2}r = 2\sqrt{2} \times 1.292$ A° $= 3.65$ A°

$\therefore$ Volume of unit cell of FCC $= a^3 = 48.78 \times 10^{-30} m^3$

No of atoms per unit cell = 4

Then volume occupied by one atom $= \frac{48.78 \times 10^{-30}}{4} = 12.2 \times 10^{-30} m^3$

% change in volume $= \frac{(12.26 - 12.2) \times 10^{-30}}{12.26 \times 10^{-30}} \times 100 = 0.5\%$

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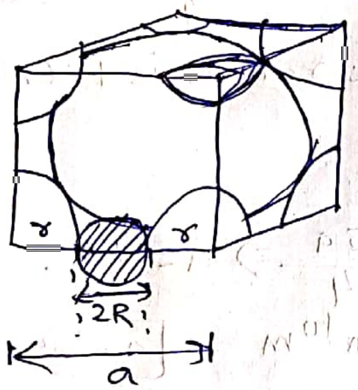
Ques-5 :- Find the minimum radius of the interstitial sphere that can fit into the void at $\left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right)$ b/w the atoms in BCC structure.

Sol^n :- In case of BCC structure $a = \frac{4r}{\sqrt{3}}$



Let, R be the radius of interstitial sphere that can fit into the void as shown in fig

From fig :- $2r + 2R = a$ or $R = \frac{(a-2r)}{2}$

$$R = \frac{\left(\frac{4}{\sqrt{3}}r - 2r\right)}{2} = \frac{2}{\sqrt{3}}r - r$$

$$R = 1.155r - r = 0.155r \Rightarrow \boxed{R = 0.155r}$$

Ques-6 :- The lattice constant for a cubic lattice is a. Deduce the spacings b/w (011), $(\bar{1}01)$ and (112) planes.

Sol^n :- For a cubic lattice, the interplanar spacing is given by $d_{hkl} = \frac{a}{\sqrt{(h^2+k^2+l^2)}}$ where h, k, l are the Miller indices.

For (011) plane, $h = 0, k = 1$ and $l = 1$

$$\therefore \quad d_{011} = \frac{a}{\sqrt{(0)^2+(1)^2+(1)^2}} = \frac{a}{\sqrt{2}}$$

Similarly, $$d_{101} = \frac{a}{\sqrt{(1)^2+(0)^2+(1)^2}} = \frac{a}{\sqrt{2}}$$

and $$d_{112} = \frac{a}{\sqrt{(1)^2+(1)^2+(2)^2}} = \frac{a}{\sqrt{6}}$$

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Ques-7:- Draw the crystal Planes with miller indices (100), (010), (001), (110), (111), (011), (002), $(\bar{1}00)$, (200), $(10\bar{1})$, $(\bar{1}\bar{1}\bar{1})$, $(1\bar{1}0)$.

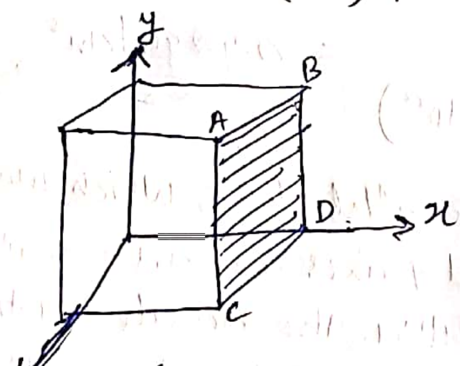
Solⁿ:- For the cubic crystal especially, the important features of miller indices are:

(i) When a plane is parallel to one of the coordinate axes, it is said to meet at infinity. in this case, $\frac{1}{\infty} = 0$ i.e. Miller index for that axis is zero.

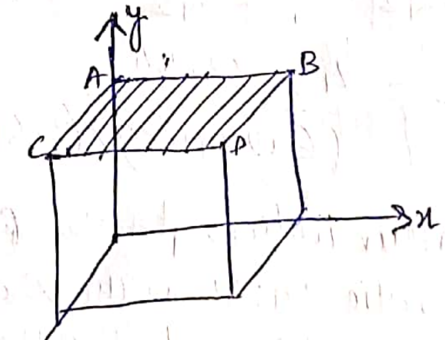
(ii) When the intercept of a plane is on the negative part of the axis, the miller index is distinguised by a bar putting directly over it.

(iii) The plane passing through the origin is defined in terms of a parallel plane having non-zero intercept. Alternately if the plane passes through the origin, the origin has to be shifted for indexing the plane.

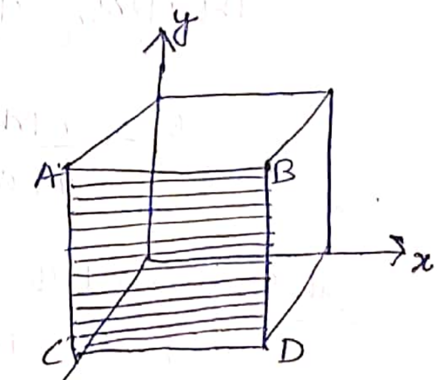
(iv) All equally spaced parallel planes have Same Miller indices i.e the miller indices do not only define a particular plane but also a set of parallel planes. for Example- The (622) planes are the same as (311) planes.



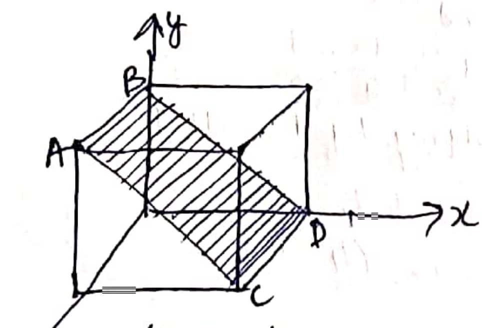
(100) plane



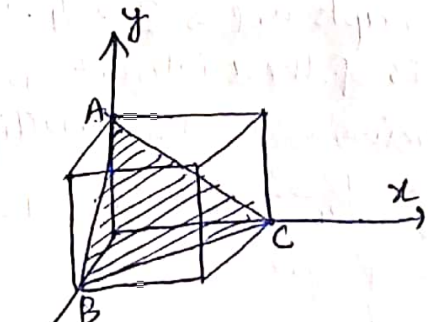
(010) plane



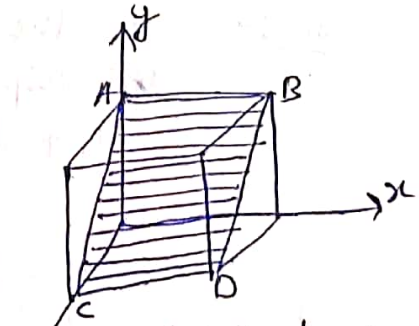
(001) plane



(110) plane



(111) plane



(011) plane

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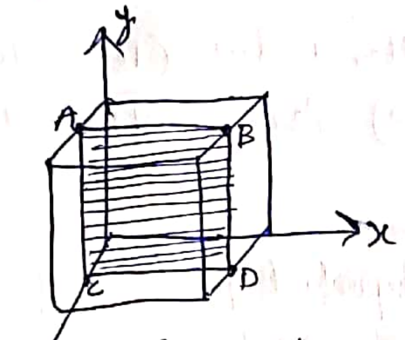
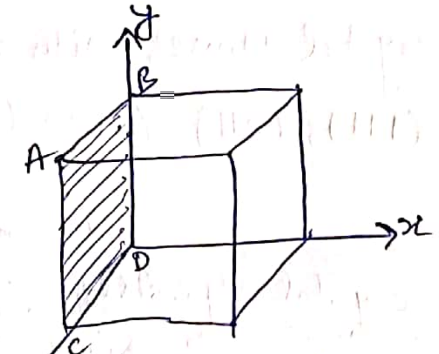
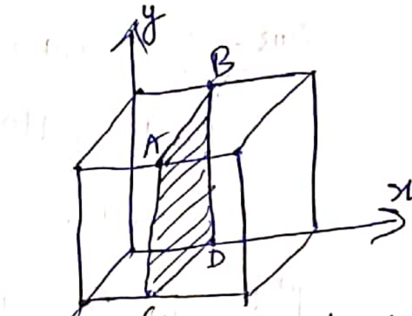


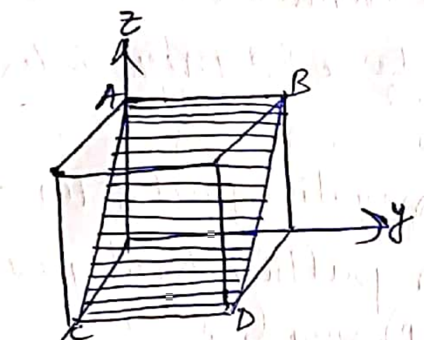
(002) plane



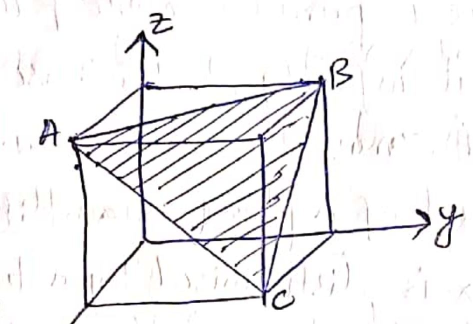
($\bar{1}00$) plane



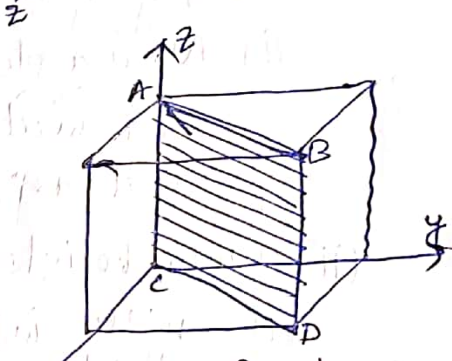
(200) plane



(101) plane



($\bar{1}\bar{1}\bar{1}$) plane



($1\bar{1}0$) plane

Ques-8 :- The atomic radius of copper is 1.278 A°. it has atomic weight 63.54 Find the density of Copper. Given $a = 4r/\sqrt{2}$.

Soln :- We Know that copper has FCC Structure.

Therefore, $a = \frac{4r}{\sqrt{2}} = \frac{4 \times 1.278}{\sqrt{2}} = 3.61\ A^\circ$

$$\rho = \frac{nM}{Na^3} = \frac{4 \times 63.54}{(6.02 \times 10^{23}) \times (3.61 \times 10^{-8})^3} = \underline{8.98\ gm/cm^3}$$

Ques-9 :- Find the Miller indices of a set of parallel planes which make interepts in the ratio 3a:4b on the X and Y axes and are parallel to Z-axis. a,b,c, being primitive vectors of the lattice. Also calculate the interplanar distance of the planes taking the lattice to be a cube with a=b=c=2A°.

Soln :- Here the intercepts are $\Rightarrow 3 : 4 : \infty$

The reciprocals of these interepts $\Rightarrow \frac{1}{3} : \frac{1}{4} : \frac{1}{\infty}$

Lcm of denominator is 12, multiplying by 12, we get.

$\frac{12}{3} : \frac{12}{4} : \frac{12}{\infty}$ or $4:3:0$

So, miller indices of given set of plane is (430).

Interplanar distance, $d_{hkl} = \frac{a}{\sqrt{h^2+k^2+l^2}}$

$\therefore d_{430} = \frac{2}{\sqrt{4^2+3^2+0^2}} = \frac{2}{\sqrt{25}} = \frac{2}{5} A^\circ$ or $\boxed{d_{430} = 0.4\ A^\circ}$

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Ques-10:- Calculate packing factor for chromium metal having BCC structure if its density = 5.96 gm/cc and atomic weight = 50.

Solⁿ:- Here $\rho = \dfrac{nM}{N a^3}$

or $5.96 = \dfrac{2 \times 50}{(6.023 \times 10^{23}) \times a^3}$

Solving for a, we get $a^3 = 1.66 \times 10^{-22}\ cm^3$ or $a = 5.49 \times 10^{-8}\ cm$

The radius of an atom in BCC structure is given by

$r = \dfrac{a\sqrt{3}}{4} = \dfrac{(5.49 \times 10^{-8}) \times \sqrt{3}}{4} = 2.38 \times 10^{-8}\ cm$

Now, packing factor $= \dfrac{\text{Volume of atoms in unit cell}}{\text{Volume of unit cell}}$

$= \dfrac{2 \times \left(\frac{4}{3}\right)\pi \times (2.3 \times 10^{-8})^3}{(1.66 \times 10^{-22})} = \underline{0.68}$

Ques-11:- Hard spheres are packed in BCC lattice in such a manner that the atom at the centre just touches the atoms present at the corners of the cube. Calculate the fraction of the BCC unit cell volume filled with hard spheres.

Solⁿ:- In BCC structure, spheres per unit cell = 2

radius of each sphere $= \dfrac{a\sqrt{3}}{4}$

$\therefore$ Volume $V = \dfrac{4}{3}\pi\left[\dfrac{a\sqrt{3}}{4}\right]^3 = \dfrac{\pi a^3\sqrt{3}}{16}$

Packing fraction (P.F.) $= \dfrac{\text{No of atoms} \times \text{volume of one atom}}{\text{Volume of unit cell}}$

$= \dfrac{2 \times \pi a^3 \sqrt{3}/16}{a^3} = \dfrac{\pi\sqrt{3}}{8} = 0.68$

$\underline{\text{P.F.} = 68\%}$. i.e. Thus 68% of BCC unit cell volume is filled with hard spheres.

Ques-12:- The first order Bragg maximum of electron diffraction in a nickel crystal ($d = 0.4086\ A^\circ$) occurred at a glancing angle of 65°. Calculate the de-Broglie wave length of the electrons and their velocity.

Solⁿ:- According to Bragg's Law $\Rightarrow$ $2d\sin\theta = n\lambda$

$\therefore \lambda = \dfrac{2d\sin\theta}{n} = \dfrac{2 \times 0.4086 \times 10^{-10} \times \sin 65^\circ}{1} = 0.7501 \times 10^{-10}\ m$

de-Broglie wave length $\lambda = \dfrac{h}{mv} \Rightarrow v = \dfrac{h}{m\lambda} = \dfrac{6.6 \times 10^{-34}}{9.1 \times 10^{-31} \times 0.75 \times 10^{-10}} = \underline{9.76 \times 10^{6}\ m/s}$

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